METROPOLIS ADVANCED HEALTHCARE SYSTEM

DEPARTMENT OF PATHOLOGY & CLINICAL LABORATORY MEDICINE

CLIA ID: 99D0874521
CAP ACCREDITED: #1478201

Patient Name: Eleanor Vance
Patient ID / MRN: PT-90482
DOB: 1972-04-12 (Age: 54 Yrs)
Gender: Female

Accession #: ACC-2026-88491
Collection Date: 2026-10-25 08:30 AM
Received Date: 2026-10-25 09:15 AM
Report Date: 2026-10-26 14:00 PM

Ordering Physician:
Dr. Marcus Vance, MD
Specimen: Whole Blood EDTA
Status: FINAL VERIFIED

CLINICAL PRESENTATION & REASON FOR TESTING

Patient presents with a 4-month history of severe fatigue, exertional dyspnea, restless leg syndrome, and cold intolerance.
Physical examination notable for marked conjunctival pallor, koilonychia, and resting tachycardia (HR 102 bpm).
Patient reports ongoing heavy menorrhagia over the preceding 6 months with suspected chronic blood loss.
Ordering evaluation for microcytic hypochromic anemia and iron deficiency.

1. COMPLETE BLOOD COUNT (CBC) & RED CELL MORPHOLOGY

Test Name	Result	Units	Reference Range	Flag
Hemoglobin (Hb)	7.8	g/dL	12.0 - 15.5	CRITICAL
Hematocrit (HCT)	24.6	%	37.0 - 48.0	LOW
RBC Count	2.95	x10^6/uL	3.80 - 5.20	LOW
Mean Corpuscular Volume (MCV)	68.2	fL	80.0 - 100.0	LOW
Mean Corpuscular Hemoglobin (MCH)	20.4	pg	27.0 - 33.0	LOW
MCHC	29.5	g/dL	32.0 - 36.0	LOW
Red Cell Distribution Width (RDW)	19.4	%	11.5 - 14.5	HIGH
White Blood Cell (WBC) Count	6.4	x10^3/uL	4.5 - 11.0	NORMAL
Platelet Count	445	x10^3/uL	150 - 400	HIGH

2. IRON STATUS & BIOCHEMICAL HEMATOLOGY PANEL

Test Name	Result	Units	Reference Range	Flag
Serum Ferritin	4.8	ng/mL	15.0 - 150.0	CRITICAL
Serum Iron	18	ug/dL	60 - 170	LOW
Total Iron Binding Capacity (TIBC)	485	ug/dL	240 - 450	HIGH
Transferrin Saturation (TSAT)	3.7	%	20.0 - 50.0	LOW

3. COMPREHENSIVE METABOLIC, RENAL & GLYCEMIC PANEL

Test Name	Result	Units	Reference Range	Flag
Glycated Hemoglobin (HbA1c)	5.3	%	< 5.7	NORMAL
Fasting Serum Glucose	88	mg/dL	70 - 99	NORMAL
Serum Creatinine	0.78	mg/dL	0.50 - 1.10	NORMAL
Blood Urea Nitrogen (BUN)	12	mg/dL	7 - 20	NORMAL
Estimated GFR (eGFR)	> 90	mL/min/1.73m2	> 60	NORMAL
Serum Sodium	140	mEq/L	136 - 145	NORMAL
Serum Potassium	4.2	mEq/L	3.5 - 5.0	NORMAL

CLINICAL INTERPRETATION & DIAGNOSTIC IMPRESSION

DIAGNOSTIC SUMMARY:

1. Severe Microcytic Hypochromic Anemia: Marked reduction in Hemoglobin (7.8 g/dL), Hematocrit (24.6%), MCV (68.2 fL), and MCH (20.4 pg), consistent with red blood cell microcytosis and hypochromia.
2. Profound Iron Deficiency: Critically depleted iron reserves evidenced by Serum Ferritin 4.8 ng/mL, Serum Iron 18 ug/dL, elevated TIBC 485 ug/dL, and transferrin saturation 3.7%.
3. Anisocytosis & Reactive Thrombocytosis: Elevated RDW (19.4%) and Platelet count (445 k/uL) reflect active bone marrow response and erythropoietic stress secondary to chronic blood loss.
4. Preserved Renal Function & Glycemic Control: Serum Creatinine (0.78 mg/dL), eGFR > 90, and HbA1c (5.3%).

RECOMMENDED CLINICAL ACTION PLAN & TESTS

1. Therapeutic Iron Repletion: Initiate intravenous iron therapy (e.g., Ferric Carboxymaltose 1000 mg IV) given the severity of anemia (Hb < 8.0 g/dL) and clinical intolerance or inadequacy of oral iron.
2. Gynecological Evaluation: Immediate pelvic ultrasound and gynecological consultation to investigate the underlying etiology of persistent menorrhagia.
3. Gastrointestinal Workup: Stool Fecal Immunochemical Test (FIT) and GI consultation for endoscopy/colonoscopy to exclude concomitant occult gastrointestinal hemorrhage.
4. Follow-up: Repeat CBC and reticulocyte count at 2 weeks post-iron administration to verify marrow response.

Electronically Verified & Authenticated by:

Dr. Sarah Jenkins, MD, FCAP

Medical Director & Chief of Hematopathology | License #: MD-883492-NY

[DIGITALLY SIGNED & ENCRYPTED]

Verification Timestamp: 2026-10-26 14:05:12 UTC